GM - 2022

Q5321094
Question Paper SL No.

2022

MATHEMATICS

[English Version]

3 Hours 15 Minutes

(First 15 minutes for reading the question paper)

Full Marks — { 90 — **For Regular Candidates**
100 — **For External Candidates**

Special credit will be given for answers which are brief and to the point.
Marks will be deducted for spelling mistakes, untidiness & bad handwriting.

(The answers of the question Nos. 1, 2, 3 are to be written at the beginning of the answer script mentioning the question numbers in the serial order. Necessary calculation and drawing must be given on the right hand side by drawing margins on the first few pages on the answer-script. Tables and Calculators of any type are not allowed. Approximate value of π may be taken as $\frac{22}{7}$, if necessary. Graph paper will be supplied, if required. Arithmetic problems may be solved by algebraic method.)

(Alternative questions of 11 are given for sightless candidates on page No. 15)

(Question number 13 on page Nos. 15 and 16 is only for external candidates)

---

1. Choose the correct option in each case from the following questions : $1 \times 6 = 6$

(i) At present the population of a village is P and if rate of increase of population per year be $2r\%$, after n years the population will be :

(a) $P\left(1 + \frac{r}{100}\right)^n$ (b) $P\left(1 + \frac{r}{50}\right)^n$

(c) $P\left(1 + \frac{r}{100}\right)^{2n}$ (d) $P\left(1 - \frac{r}{100}\right)^n$

(ii) Fatima, Shreya and Smita started a business by investing total Rs. 6,000. After a year Fatima, Shreya and Smita get profit share of Rs. 50, Rs. 100 and Rs. 150 respectively. Smita invested in this business :

(a) Rs. 1,000 (b) Rs. 2,000

(c) Rs. 3,000 (d) Rs. 4,000

(iii) If $A : B = 2 : 3$, $B : C = 5 : 8$, $C : D = 6 : 7$, then $A : D =$

(a) $2 : 7$ (b) $7 : 2$

(c) $5 : 8$ (d) $5 : 14$

(iv) 'O' is the centre of a circle and PQ is a diameter. R is a point on the circle such that $PR = RQ$, then the value of $\angle RPQ$:

(a) 30° (b) 90°

(c) 60° (d) 45°

(v) If two circles do not intersect or touch each other, then the maximum number of common tangent/tangents is/are :

(a) 2 (b) 1

(c) 3 (d) 4

(vi) The volume of a solid sphere having radius $2r$ units is :

(a) $\frac{32\pi r^3}{3}$ cubic unit (b) $\frac{16\pi r^3}{3}$ cubic unit

(c) $\frac{8\pi r^3}{3}$ cubic unit (d) $\frac{64\pi r^3}{3}$ cubic unit

2. Fill up the blanks (any **five**) : $1 \times 5 = 5$

(i) The annual rate of compound interest is $r\%$ and if the first year principal is P, then 2nd year principal is .....

(ii) $7\sqrt{11}$ is an ..... number.

(iii) If radius of a sphere is r and volume v, then $v \propto$ .....

(iv) Two triangles are similar if their corresponding sides are .....

(v) If the opposite angles of a quadrilateral be supplementary, then the vertices of the quadrilateral will be .....

(vi) If the length, breadth and height of a rectangular parallelepiped are equal, then the special name of the solid is .....

3. Write True or False (any **five**) : $1 \times 5 = 5$

(i) At least 3 persons are needed in partnership business.

(ii) The relation between principal and amount is principal < amount.

(iii) The two roots of the equation $x^2 = 100$ are ± 10.

(iv) If a and b are in inverse variation, then $\frac{a}{b} = \text{constant}$.

(v) Two concentric circles have only one common tangent.

(vi) The height, radius and slant height of a right circular cone are always the three sides of a right angled triangle.

4. Answer the following questions (any **ten**) : $2 \times 10 = 20$

(i) The annual interest is $\frac{1}{16}$ part of its principal, then determine the interest of Rs. 690 of 8 months.

(ii) The present population is 13,310. If the population be 17,280 after 3 years what will be the rate of increase?

(iii) The ratio of capitals of A, B, C is $\frac{1}{x} : \frac{1}{y} : \frac{1}{z}$, after a year there was a loss of Rs. z. Calculate the loss of C.

(iv) Find out the ratio of the sum and the product of two roots of the equation $7x^2 - 66x + 27 = 0$.

(v) Rationalise the surds of denominator : $\frac{12}{\sqrt{15}-3}$

(vi) The radius of the circle with the centre 'O' is 13 cm and a chord AB with the length of 10 cm on it. Calculate the distance of the chord AB from the centre of the circle.

(vii) AOB is a diameter of a circle whose centre is O. The point C lies on the circle. If $\angle \text{OBC} = 60^\circ$, find the value of $\angle \text{OCA}$.

(viii) A circle with the centre 'O'. A point P is 26 cm away from the centre of the circle and the length of the tangent drawn from the point P to the circle is 10 cm. Calculate the length of radius of the circle.

(ix) $\text{DE} \parallel \text{BC}$ of ΔABC where D and E are two points on AB and AC respectively. If $\text{AD} = 5$ cm , $\text{DB} = 6$ cm and $\text{AE} = 7.5$ cm. Calculate the length of AC.

(x) If the height of two right circular cylinders are in the ratio of $1 : 2$ and perimeters of the base are in the ratio of $3 : 4$. Find the ratio of their volumes.

(xi) If the length of radius of a sphere is increased by 50%, find how much percent will be increased of its curved surface area.

(xii) The length of diagonal of a cube is $4\sqrt{3}$ cm. Calculate the total surface area of the cube.

5. Answer any **two** questions : $5 \times 2 = 10$

(i) At the same rate of simple interest in percent per annum, if a principal becomes the amount of Rs. 7,100 in 7 years and of Rs. 6,200 in 4 years, determine the principal and rate of simple interest in percent per annum.

(ii) Three friends have started a business by investing Rs. 8,000, Rs. 10,000 and Rs. 12,000 respectively. They also took an amount as bank loan. At the end of one year they made a profit of Rs. 13,400. After paying the annual bank installment of Rs. 5,000 they divided the remaining money of the profit among themselves in the ratio of their capitals. Calculate the profit share of each.

(iii) Calculate the difference between compound interest and simple interest on Rs. 20,000 for 2 years at 5% per annum.

6. Answer any **two** questions : $3 \times 2 = 6$

(i) Solve : $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}, x \neq 0, -(a+b)$

(ii) Form the quadratic equation whose roots are -4 and 3.

(iii) If $m + \frac{1}{m} = \sqrt{3}$, then find the value of

(a) $m^2 + \frac{1}{m^2}$ and (b) $m^3 + \frac{1}{m^3}$.

7. Answer any **two** questions : $3 \times 2 = 6$

(i) Find the simplest value of : $\frac{\sqrt{5}}{\sqrt{3}+\sqrt{2}} - \frac{3\sqrt{3}}{\sqrt{2}+\sqrt{5}} + \frac{2\sqrt{2}}{\sqrt{3}+\sqrt{5}}$

(ii) If $a = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $ab = 1$, calculate $\left(\frac{a}{b} + \frac{b}{a}\right)$.

(iii) If 15 farmers can cultivate 18 bighas of land in 5 days, determine by using theory of variation the number of days required by 10 farmers to cultivate 12 bighas of land.

8. Answer any **one** question : $3 \times 1 = 3$

(i) If $a : b = b : c$, then prove that : $\frac{abc(a+b+c)^3}{(ab+bc+ca)^3} = 1$

(ii) If $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = 1$, then find the value of : $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$

9. Answer any **one** question : $5 \times 1 = 5$

(i) Prove that the opposite angles of any cyclic quadrilateral are supplementary.

(ii) Prove that if a perpendicular is drawn from the centre of a circle on a chord which is not a diameter, then that perpendicular bisects the chord.

10. Answer any **one** question : $3 \times 1 = 3$

(i) ABCD is a cyclic quadrilateral. The chord DE is the external bisector of $\angle$BDC. Prove that AE (or extended AE) is the external bisector of $\angle$BAC.

(ii) The two chords AB and CD of a circle with centre O when produced intersect each other at the point P. Prove that $\angle AOC - \angle BOD = 2\angle BPC$.

11. Answer any **one** question : $5 \times 1 = 5$

(i) Draw a right-angled triangle whose sides containing the right angle are 4 cm and 8 cm in length. Draw the circumcircle of the triangle. (Only traces of construction are required)

(ii) Construct a circle of radius 3 cm. Take a point outside the circle and at a distance of 6 cm from the centre. From that point, draw a tangent to the circle. (Only traces of construction are required)

12. Answer any **four** questions : $4 \times 4 = 16$

(i) A rectangular cistern 2.1 m long and 1.5 m wide is half-filled with water. If 630 litres of water is poured into the cistern, calculate how much the level of water will rise.

(ii) The height of a right circular cylinder is twice its radius. If the height were 6 times of its radius, the volume of the cylinder would be 539 cu. dm more. Find the height of the cylinder.

(iii) A conical tent can accommodate 11 persons. Each person requires 4 sq. m space on the ground and 20 cu. m of air for breathing. Find the height of the tent constructed for these 11 persons.

(iv) Find how many solid spherical bullets of 1 cm radius can be made by melting a solid iron sphere of 8 cm radius.

(v) The inner length, breadth and height of a tea-box are 7.5 dm, 6 dm and 5.4 dm respectively. The weight of the box filled with tea is 52 kg 350 g, but the weight of the empty box is 3.75 kg. Find the weight of 1 cu. dm of tea.

learningscience.co.in